Unconfounded but inflated causal estimates

Mathematical derivation

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I Setting

I. 1 Objectives

The objective of my article is to compare "biases" resulting from exaggeration in causal identification strategies on the one hand and from confounders in "naive" regression analysis on the other hand. This document aims to provide a mathematical derivation to illustrate this trade-off. Such a derivation may provide some intuition regarding the various mechanisms at play. It does not aim in any way to provide a general proof but merely a specific one, tied to the setting considered. To do so, I build on a simple and commonly used framework. While I keep the setting as general as possible, I have to make certain assumptions, in particular regarding the data generating process considered.

The objective in this mathematical derivation is to compare the exaggeration ratio for a naive regression with omitted variable bias (OVB) to that of the regression in which we control for the omitted variable (CTRL), of an Instrumental Variable (IV) model in which we instrument the variable affected by the omitted variable and of a reduced form (RED) estimation in which we directly regress the outcome on the instrument. Roughly, we aim to derive the conditions under which the exaggeration due to the loss in precision is greater than the omitted variable bias. The objective is to compare:

$$\mathbb{E}\left[\hat{\beta}_{\text{OVB}} |\ \beta_1, \sigma_{\text{OVB}}, \hat{\beta}_{\text{OVB}} > z_\alpha \sigma_{\text{OVB}}\right] \qquad \text{and} \qquad \mathbb{E}\left[\hat{\beta}_{..} |\ \beta_1, \sigma_{..}, \hat{\beta}_{..} > z_\alpha \sigma_{..}\right]$$

where .. stands for the controlled regression (CTRL), the IV (IV) and the reduced form (RED). β_1 represents the true effect of interest, $\hat{\beta}_{..}$ its estimate, $\sigma_{..}$ the standard error of the estimate and z_α the threshold value of a two-sided test of size α. In all of my analyses, I consider tests of size 5%. One can thus think of z_α as being equal to 1.96.

In this document, I first describe the data generating process. I then derive a general equation for the exaggeration of a normally distributed estimator and discus how it evolves with the mean and standard deviation of the estimator. In a third section, I build an hand-wavy illustration of the confounding / exaggeration trade-off when we control for the omitted variable. Since most causal identification strategies can be interpreted as "controlling" for confounders (*e.g.* the control function (CF) approach for the IV), the CTRL case neatly summarizes the intuition for all strategies. In the last section, I build more formal proofs on the existence of the trade-off for the control approach, the IV and event-study estimated via 2SLS and the event-study estimated as a reduced form. For each strategy, I analyze how exaggeration evolves as a function of the parameters of interests. I also discuss how for each strategy, the main parameter of interest is linked the proportion of the variance in the explanatory variable of interest retained for identification.

I. 2 Data generating process

Consider the usual linear regression model to with an omitted variable. We limit the number of covariates considered to a constant, an observed variable and an omitted variable. For all individual $i \in \{1, ..., n\}$, assume that:

$$y_i = \beta_0 + \beta_1 x_i + \delta w_i + u_i \tag{1}$$

where $x_i \in \mathbb{R}$ is the observed explanatory variable of interest, $y_i \in \mathbb{R}$ the observed outcome of interest, $w_i \in \mathbb{R}$ an unobserved omitted variable, $u_i \in \mathbb{R}$ an unobserved error term. $(\beta_0, \beta_1, \delta) \in \mathbb{R}^3$ are unknown parameters. β_1 is the parameter of interest. Treatment effects are homogeneous ($\forall i \in \{1, ..., n\},\ \beta_{1i} = \beta_1 = \text{cst}$). This equation can be rewritten as:

$$y_i = \mathrm{x}_i' \boldsymbol{\beta} + \delta w_i + u_i$$

with $\boldsymbol{\beta} \in (\beta_0, \beta_1)'$ and $\mathrm{x}_i = (1, x_i)'$. For readability, we introduce the usual vector notation such that for instance $y = (y_1, ..., y_n)'$.

We further assume that the usual OLS assumptions hold:
— $(y_i, \mathrm{x}_i, w_i, u_i)$ are iid,
— Finite second moments, *ie* $\mathbb{E}[y_i^2] < \infty$, $\mathbb{E}[w_i^2] < \infty$, $\mathbb{E}[x_i^2] < \infty$,
— Positive-definiteness of $\mathbb{E}[\mathrm{x}_i \mathrm{x}_i']$,
— u_i is conditional mean-zero, *ie* $\mathbb{E}[u_i | x_i, w_i] = 0$, and is uncorrelated with x_i and w_i, *ie* $\mathbb{E}[x_i u_i] = 0$ and $\mathbb{E}[w_i u_i] = 0$.

Note that conditional mean-zero implies that $\mathbb{E}[u_i | x_i] = 0$ (since $\mathbb{E}[u_i | x_i] = \mathbb{E}[\mathbb{E}[u_i | x_i, w_i | x_i]] = \mathbb{E}[0 | x_i] = 0$). We further assume homoskedasticity such that $\mathbb{E}[u_i^2 | x_i, w_i] = \sigma_u^2$ is constant. It implies that $\mathbb{E}[u_i^2 | x_i] = \sigma_u^2$ (since $\mathbb{E}[u_i^2 | x_i] = \mathbb{E}[\mathbb{E}[u_i^2 | x_i, w_i] | x_i] = \mathbb{E}[\sigma_u^2 | x_i] = \sigma_u^2$).

We however assume that w_i is unobserved, correlated with x_i and that $\delta \neq 0$. To simplify the derivations, we further assume that the unobserved variable is centred, *ie* $\mathbb{E}[w_i] = 0$. We also assume that the variance of the component of w_i that is orthogonal to x_i (denoted $w_i^\perp$) does not vary with x_i. Thus, $\mathrm{Var}(w_i^\perp | x_i) = \mathrm{Var}(w_i^\perp)$.

We further assume that there exists a valid instrumental variable z_i for x_i. Let $\mathrm{z} = (1, z_i)' \in \mathbb{R}^2$. Note that the model is just-identified. Since the instrument is valid, it satisfies exogeneity, *ie* $\mathbb{E}[\mathrm{z}_i u_i] = 0$ and $\mathbb{E}[z_i w_i] = 0$, relevance, *ie* $\mathrm{rank}(\mathbb{E}[\mathrm{z}_i \mathrm{x}_i']) = 2$, and positive-definiteness of $\mathbb{E}[\mathrm{z}_i \mathrm{z}_i']$. To simplify interpretation, we further assume a data generating process for x_i:

$$x_i = \pi_0 + \pi_1 z_i + \gamma w_i + \epsilon_i \tag{2}$$

where $(\pi_0, \pi_1, \gamma) \in \mathbb{R}^3$ are unknown parameters. We assume that ϵ_i is uncorrelated with z_i and w_i, *ie* $\mathbb{E}[z_i \epsilon_i] = 0$ and $\mathbb{E}[w_i \epsilon_i] = 0$. Note that, since x_i and w_i are correlated, $\gamma \neq 0$. We also assume homoskedasticity for this term, such that $\mathbb{E}[\epsilon_i^2 | z_i, w_i] = \sigma_\epsilon^2$ is constant. It implies that $\mathbb{E}[\epsilon_i^2 | z_i] = \sigma_\epsilon^2$ (since $\mathbb{E}[\epsilon_i^2 | z_i] = \mathbb{E}[\mathbb{E}[\epsilon_i^2 | z_i, w_i] | z_i] = \mathbb{E}[\sigma_\epsilon^2 | z_i] = \sigma_\epsilon^2$).

The following Directed Acyclic Graph (DAG) sums up the relationships between the variables. It also helps underline the independence, or not, of some variables.

In the reduced form case, we consider an analogous DGP to that of the IV. Yet, since we will directly regress y on z, the resulting estimand will be different. To make them comparable, for the reduced form case, in equation 2, we set π_1 to 1. An increase of 1 in z causes an increase of β_1 in y and the estimands become comparable. Further, we focus on the case where z is an exogenous discrete

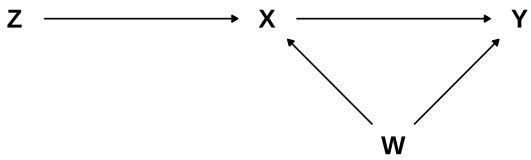

Figure 1 – DAG of the data generating process

shock. For clarity, we denote this discrete shock $\mathrm{d}_i = (1, d_i)'$. All other assumptions and features of the DGP remain unchanged. Under these assumptions, plugging equation 2 into equation 1 yields:

$$y_i = \underbrace{\beta_0 + \beta_1\pi_0}_{b_0} + \beta_1 d_i + \underbrace{(\delta + \beta_1\gamma)}_{\kappa} w_i + \underbrace{u_i + \beta_1\epsilon_i}_{\nu_i}$$

The second component of the parameter vector associated with d_i is thus identical to the second component of $\boldsymbol{\beta}$. For the sake of clarity, we also assume that $\pi_0 = 0$ such that $b_0 = \beta_0$ and thus $(b_0, \beta_1) = \boldsymbol{\beta}'$. This assumption is not necessary but clarifies the analysis. Dropping does not change the results for the second component of $\boldsymbol{\beta}$, the component of interest. Thus, in the reduced form case, we have:

$$y_i = \mathrm{d}_i'\boldsymbol{\beta} + \kappa w_i + \nu_i \tag{3}$$

I. 3 Summary of the key assumptions

To ease the use of assumptions in the following sections, we summarize some that we will explicitly mobilize:

— $y_i = \mathrm{x}_i'\boldsymbol{\beta} + \delta w_i + u_i$ ($y_i = \mathrm{d}_i'\boldsymbol{\beta} + \kappa w_i + \nu_i$ in the reduced form case).
— $x_i = \pi_0 + \pi_1 z_i + \gamma w_i + \epsilon_i$ (with $x_i = 1 \cdot d_i + \gamma w_i + \epsilon_i$ in the reduced form case).
— $\mathbb{E}[u_i|x_i, w_i] = 0$ and $\mathbb{E}[u_i|x_i] = 0$
— $\mathbb{E}[u_i^2|x_i, w_i] = \sigma_u^2$ and $\mathbb{E}[u_i^2|x_i] = \sigma_u^2$
— $\mathbb{E}[w_i] = 0$
— $\mathbb{E}[z_i u_i] = 0$ and $\mathbb{E}[z_i w_i] = 0$
— $\mathbb{E}[z_i \epsilon_i] = 0$ and $\mathbb{E}[w_i \epsilon_i] = 0$
— $\mathbb{E}[\epsilon_i^2|z_i, w_i] = \sigma_\epsilon^2$ and $\mathbb{E}[\epsilon_i^2|z_i] = \sigma_\epsilon^2$
— $\mathrm{Var}(w_i^\perp|x_i) = \mathrm{Var}(w_i^\perp)$
— $\beta_1 > 0$ and $\delta\gamma > 0$ (which implies that the OVB is an upward bias).

I. 4 The models to estimate

We consider 3 models:

1. A "naive" regression of y on x (with w omitted). Since, we do not observe w, we consider the projection of y on X only:

$$y = X\boldsymbol{\beta}_{\text{OVB}} + v \tag{4}$$

where $v = (v_1, ..., v_n)' \in \mathbb{R}^n$ and by definition of the projection, $\mathbb{E}[X'v] = 0$. Note that we use capital letters to denote matrices (for instance $X = (\mathrm{x}_1', ..., \mathrm{x}_n')'$, with $\forall i, \mathrm{x}_i' = (1, x_i)$).

2. An IV regression in which we regress y on x instrumented by z. More precisely, we instrument $\mathrm{x}_i = (1, x_i)'$ with $\mathrm{z}_i = (1, z_i)'$. We thus consider a just-identified case and $\hat{\boldsymbol{\beta}}_{\text{2SLS}} = \hat{\boldsymbol{\beta}}_{\text{IV}}$.
3. A reduced form equation in which we regress y on the exogenous shock D:

$$y = D\boldsymbol{\beta}_{\text{RED}} + e$$

where $e = (e_1, ..., e_n)' \in \mathbb{R}^n$ and by the exogeneity assumption $\mathbb{E}[D'e] = 0$.

II Exaggeration ratio and its properties

I first compute the exaggeration ratio for a normally distributed estimator and study its general properties. This then enables me to illustrate the intuition behind the link between imprecision and exaggeration.

II. 1 A general expression for the exaggeration ratio

Lemma 1. *For an estimator $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2)$ of a true effect of magnitude $\beta_1 > 0$ and such that $\beta = \beta_1 + b$, the exaggeration ratio E defined as the ratio expected value of the estimate conditional on the estimate being statistically significant over the true effect is:*

$$E(\beta, \sigma, \beta_1, z_\alpha) = \frac{\mathbb{E}\left[\hat{\beta} |\ \beta_1, \sigma, \hat{\beta} > z_\alpha \sigma\right]}{\beta_1} = 1 + \frac{b}{\beta_1} + \frac{\sigma}{\beta_1}\lambda\left(\frac{\beta_1 + b}{\sigma} - z_\alpha\right) \tag{5}$$

Note that Gelman & Carlin (2014) consider the expected value of the *absolute value* of the estimate: $\mathbb{E}\left[|\hat{\beta}|| \ \beta_1, \sigma, \hat{\beta} > z_\alpha \sigma\right] / \beta_1$. To simplify the derivation, I abstract from the absolute value. My exaggeration ratio therefore depends on the well-known expected value of a truncated normal distribution instead of on the expected value of a truncated *folded* normal distribution.

Also note that, since the objective of this project is mostly to analyze when the exaggeration is greater than the OVB, it makes sense to focus on cases in which they bias the estimate in the same direction *i.e.* β_1 and b of the same sign. I focus on the case where the true effect is positive and the OVB upward, *i.e.*, $\beta_1 > 0$ and $b \geq 0$. The reasoning for negative true effect and bias are is analogous but with signs flipped.

Finally, note that in this document, I refer to E as the "exaggeration ratio" and $\mathbb{E}\left[\hat{\beta} |\ \beta_1, \sigma, \hat{\beta} > z_\alpha \sigma\right]$ as "exaggeration" to avoid using the two terms interchangeably.

Proof. This exaggeration ratio is can be derived directly using properties of a truncated normal or directly using the expression of the inverse Mills ratio. Yet, since this document is a thorough description of the analysis, I derive this expression analytically.

Since $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2)$, its support is $\mathbb{R}$ and pdf is $f_{\hat{\beta}} : x \mapsto \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\beta)^2}{2\sigma^2}}$. Thus, by definition of the

conditional expectation,

$$\mathbb{E}\left[\hat{\beta}|\ \beta_1, \sigma, \hat{\beta} > z_\alpha \sigma\right] = \frac{1}{\Pr(\hat{\beta} > z_\alpha \sigma)} \int_{z_\alpha \sigma}^{\infty} x \times f_{\hat{\beta}}(x) \mathrm{d}x$$

Now,

$$\begin{aligned}
\Pr(\hat{\beta} > z_\alpha \sigma) &= \int_{z_\alpha \sigma}^{\infty} f_{\hat{\beta}}(x) \mathrm{d}x \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{z_\alpha \sigma}^{\infty} e^{-\frac{(x-\beta)^2}{2\sigma^2}} \mathrm{d}x \\
&= \frac{1}{\sqrt{2\pi}} \int_{z_\alpha - \frac{\beta}{\sigma}}^{\infty} e^{-\frac{t^2}{2}} \mathrm{d}t && \text{by the change of variable } x = \sigma t + \beta \\
&= 1 - \Phi\left(z_\alpha - \frac{\beta}{\sigma}\right) && \text{with } \Phi \text{ the cdf of the std. normal} \\
&= \Phi\left(\frac{\beta}{\sigma} - z_\alpha\right) && \text{since } \forall x \in \mathbb{R},\ \Phi(-x) = 1 - \Phi(x)
\end{aligned}$$

and,

$$\begin{aligned}
\int_{z_\alpha \sigma}^{\infty} x \times f_{\hat{\beta}}(x) \mathrm{d}x &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{z_\alpha \sigma}^{\infty} x e^{-\frac{(x-\beta)^2}{2\sigma^2}} \mathrm{d}x \\
&= \frac{1}{\sqrt{2\pi}} \int_{z_\alpha - \frac{\beta}{\sigma}}^{\infty} (\sigma t + \beta) e^{-\frac{t^2}{2}} \mathrm{d}t && \text{by the same change of variable} \\
&= \beta \times \frac{1}{\sqrt{2\pi}} \int_{z_\alpha - \frac{\beta}{\sigma}}^{\infty} e^{-\frac{t^2}{2}} \mathrm{d}t + \sigma \frac{1}{\sqrt{2\pi}} \int_{z_\alpha - \frac{\beta}{\sigma}}^{\infty} t e^{-\frac{t^2}{2}} \mathrm{d}t \\
&= \beta \left(1 - \Phi\left(z_\alpha - \frac{\beta}{\sigma}\right)\right) + \sigma \frac{1}{\sqrt{2\pi}} \left[-e^{-\frac{t^2}{2}}\right]_{z_\alpha - \frac{\beta}{\sigma}}^{\infty} \\
&= \beta \Phi\left(\frac{\beta}{\sigma} - z_\alpha\right) + \sigma \frac{1}{\sqrt{2\pi}} e^{-\frac{(z_\alpha - \beta)^2}{2\sigma^2}} \\
&= \beta \Phi\left(\frac{\beta}{\sigma} - z_\alpha\right) + \sigma \phi\left(z_\alpha - \frac{\beta}{\sigma}\right) && \text{with } \phi \text{ the pdf of the std. norm.} \\
&= \beta \Phi\left(\frac{\beta}{\sigma} - z_\alpha\right) + \sigma \phi\left(\frac{\beta}{\sigma} - z_\alpha\right) && \text{since } \forall x \in \mathbb{R},\ \phi(-x) = \phi(x)
\end{aligned}$$

Thus, we have,

$$E(\beta, \sigma, \beta_1, z_\alpha) = \frac{\beta}{\beta_1} + \frac{\sigma}{\beta_1} \frac{\phi\left(\frac{\beta}{\sigma} - z_\alpha\right)}{\Phi\left(\frac{\beta}{\sigma} - z_\alpha\right)}$$

where ϕ and Φ are respectively the pdf and the cdf of the standard normal distribution. We then introduce the inverse Mills ratio, the function $\lambda : x \mapsto \phi(x)/\Phi(x)$. We also consider an estimator whose mean can be decomposed in $\beta = \beta_1 + b$, with β_1 the true effect and b a bias independent of β_1. We

can then write the exaggeration ratio as:

$$\boxed{E(\beta, \sigma, \beta_1, z_\alpha) = 1 + \frac{b}{\beta_1} + \frac{\sigma}{\beta_1}\lambda\left(\frac{\beta_1 + b}{\sigma} - z_\alpha\right)}$$

□

Another important metric in this paper is statistical power. It defined as the probability of rejecting the null of no effect when it is actually true. Hence, for a usual two-sided t-test with $n \gg k$, with k the number of covariates, the statistical power p is given by:

$$\begin{aligned} p(\beta, \sigma, \beta_1, z_\alpha) &= \Pr(\hat{\beta} < -z_\alpha\sigma) + \Pr(\hat{\beta} > z_\alpha\sigma) \\ &= \Pr\left(\frac{\hat{\beta} - \beta}{\sigma} < -z_\alpha - \frac{\beta}{\sigma}\right) + \Pr\left(\frac{\hat{\beta} - \beta}{\sigma} > z_\alpha - \frac{\beta}{\sigma}\right) \\ &= \Phi\left(-z_\alpha - \frac{\beta}{\sigma}\right) + 1 - \Phi\left(z_\alpha - \frac{\beta}{\sigma}\right) \qquad \text{since } \frac{\hat{\beta} - \beta}{\sigma} \sim \mathcal{N}(0, 1) \end{aligned}$$

And thus, since $\forall x \in \mathbb{R},\ \Phi(-x) = 1 - \Phi(x)$,

$$\boxed{p(\beta, \sigma, \beta_1, z_\alpha) = 1 - \Phi\left(\frac{\beta}{\sigma} + z_\alpha\right) + \Phi\left(\frac{\beta}{\sigma} - z_\alpha\right)} \tag{6}$$

II. 2 Evolution of the exaggeration ratio with σ, β and b

Corollary 1. *For $\beta_1 > 0$ and $b > 0$, the exaggeration ratio increases with σ, the standard error of the estimator and decreases with β_1, the true effect size. It thus decreases with the Signal to Noise Ratio (SNR) of the estimator $\frac{\beta_1}{\sigma}$.*

Proof. From equation 5, $\forall(\beta, \sigma, \beta_1, z_\alpha) \in \mathbb{R}_+^4$,

$$\frac{\partial E}{\partial \sigma}(\beta, \sigma, \beta_1, z_\alpha) = \frac{1}{\beta_1}\lambda\left(\frac{\beta_1 + b}{\sigma} - z_\alpha\right) - \frac{\beta_1 + b}{\sigma}\lambda'\left(\frac{\beta_1 + b}{\sigma} - z_\alpha\right)$$

Since the inverse Mills ratio being a positive, decreasing function, $\forall x \in \mathbb{R}, \lambda(x) > 0$ and $\lambda'(x) < 0$. Thus, $\forall(\beta, \sigma, \beta_1, z_\alpha) \in \mathbb{R}_+^4$, $\frac{1}{\beta_1}\lambda\left(\frac{\beta_1+b}{\sigma} - z_\alpha\right) > 0$ and $-\frac{\beta_1+b}{\sigma}\lambda'\left(\frac{\beta_1+b}{\sigma} - z_\alpha\right) > 0$ and $\dfrac{\partial E}{\partial \sigma} > 0$. <u>The exaggeration ratio is an increasing function of σ, the variance of the estimator</u>.

Similarly, from equation 5, $\forall(\beta, \sigma, \beta_1, z_\alpha) \in \mathbb{R}_+^4$,

$$\frac{\partial E}{\partial \beta_1}(\beta, \sigma, \beta_1, z_\alpha) = -\frac{b}{\beta_1^2} + \frac{1}{\beta_1}\lambda'\left(\frac{\beta_1 + b}{\sigma} - z_\alpha\right) - \frac{\sigma}{\beta_1^2}\lambda\left(\frac{\beta_1 + b}{\sigma} - z_\alpha\right)$$

$\forall(\beta, \sigma, \beta_1, z_\alpha) \in \mathbb{R}_+^4$, each term of this partial derivative is negative and thus $\dfrac{\partial E}{\partial \sigma} < 0$. <u>The exaggeration ratio is a decreasing function of β_1, the true effect size</u>.

Combining these two results implies that the exaggeration ratio is a decreasing function of β_1/σ, the signal to noise ratio. □

Note that these two results have been proven by Lu et al. (2019) and Zwet & Cator (2021) in the case of an exaggeration defined as in Gelman & Carlin (2014), *i.e.*, as the expected value of the absolute value of the point estimate.

Figure 2 gives an intuition for the relationship between the precision of an estimator and exaggeration by ploting the distribution of two unbiased estimators (that therefore have identical means) but different variances and distinguishing between estimates that are and are not significant.

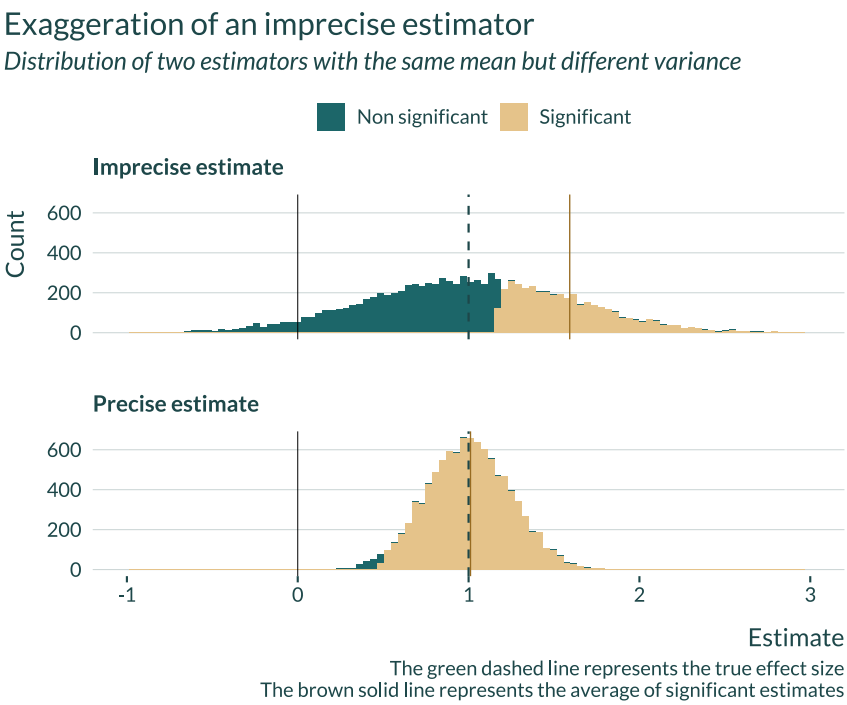

Figure 2 – Intuition for exaggeration of imprecise estimates

In the case of the precise estimator, the mean of statistically significant estimates is almost equal to the true effect. The estimator is unbiased and almost all estimates are significant. However, in the case of the imprecise estimator, only a sub-sample of the estimates are significant: those that are far away enough from 0 (at least 1.96 s.d.). As a consequence, the mean of statistically significant estimates is different from the true effect, even if the estimator is unbiased. This hints to the trade-off between biased estimates from naive regressions and less precise causal inference estimates.

III Intuition for controls and exaggeration

In this section, I give the intuition for why controlling for an omitted variable can actually cause exaggeration, depending on the characteristics of the variable one controls for. I then, based on this result, I discuss how using causal identification strategies can increase exaggeration.

III. 1 Controlling can increase exaggeration

III. 1.1 Controlling can increase variance...

In section II, we have seen that the exaggeration increases with the variance of a normally distributed estimator. Here, we will discuss under which conditions controlling for an omitted variable can increase the variance of the estimator and thus exaggeration. To do so, consider the DGP described in section I. 2 and the two following regression models:

1. A "naive" regression of y on x (with w omitted):

$$y = \beta_0^{\text{OVB}} + \beta_1^{\text{OVB}} x + u^{\text{OVB}} \tag{7}$$

2. The controlled regression of y on x and w:

$$y = \beta_0^{\text{CTRL}} + \beta_1^{\text{CTRL}} x + \delta^{\text{CTRL}} w + u^{\text{CTRL}} \tag{8}$$

Since we omitted w in equation 7, we can show the usual result such know that $\beta_1^{\text{OVB}} \neq \beta_1$ [1]. Similarly, we can easily show that $\beta_1^{\text{CTRL}} = \beta_1$. We know that the variance of β_1^{OVB} is given by:

$$\text{Var}(\hat{\beta}_1^{\text{OVB}}) = \sigma^2_{\text{OVB}} = \frac{\text{Var}(u^{\text{OVB}})}{n\ \text{Var}(x)} = \boxed{\sigma^2_{\text{OVB}} = \frac{\text{Var}(y^{\perp x})}{n\ \text{Var}(x)}}$$

where $y^{\perp x}$ is the component of y that is orthogonal to x. Using similar notation, and Frisch-Waugh-Lovell theorem, the variance of β_1^{CTRL} is:

$$\text{Var}(\hat{\beta}_1^{\text{CTRL}}) = \sigma^2_{\text{CTRL}} = \frac{\text{Var}(u^{\text{CTRL}})}{n\ \text{Var}(x^{\perp w})} = \boxed{\sigma^2_{\text{CTRL}} = \frac{\text{Var}(y^{\perp x,w})}{n\ \text{Var}(x^{\perp w})}}$$

Based on these two equations, we can see that, under certain conditions, the variance of the estimator is greater when we control for the omitted variable than when we do not:

$$\begin{aligned}
\sigma^2_{\text{CTRL}} > \sigma^2_{\text{OVB}} \quad &\Leftrightarrow \quad \frac{\text{Var}(y^{\perp x,w})}{n\ \text{Var}(x^{\perp w})} > \frac{\text{Var}(y^{\perp x})}{n\ \text{Var}(x)} \\
&\Leftrightarrow \quad \frac{\text{Var}(y^{\perp x,w})}{\text{Var}(y^{\perp x})} > \frac{\text{Var}(x^{\perp w})}{\text{Var}(x)} \\
&\Leftrightarrow \quad 1 - R^2_{y^{\perp x} \sim w} > 1 - R^2_{x \sim w} \\
&\Leftrightarrow \quad R^2_{y^{\perp x} \sim w} < R^2_{x \sim w}
\end{aligned}$$

where $R^2_{x \sim w}$ is the R^2 or coefficient of determination of the regression of x on w. This notation follows Cinelli & Hazlett (2020). $(1 - R^2_{x \sim w})$ and $\frac{\text{Var}(x^{\perp w})}{\text{Var}(x)}$ thus both represent the Fraction of the Variance in x Unexplained (FVU) by w.

Hence, <u>if the omitted variable explains a larger part of the variance of explanatory variable than</u>

1. We do so in equation 12, noting that in our case, the second componnent of $\mathbb{E}[X'X]^{-1}\mathbb{E}[X'w]$ is the coefficient for x in the regression of w on x. Denoting it $\tilde{\gamma}$, we have $\beta_1^{\text{OVB}} = \beta_1 + \delta\tilde{\gamma} \neq \beta_1$. Note that, we have $\tilde{\gamma} = \frac{\text{cov}(w,x)}{\sigma_x^2} = \frac{\text{cov}(w,x)}{\sigma_w^2}\frac{\sigma_w^2}{\sigma_x^2} = \frac{\gamma\sigma_w^2}{\sigma_x^2}$ which yields the same result as in section IV. 1, *i.e.* $\beta_1^{\text{OVB}} = \beta_1 + \delta\gamma\frac{\sigma_w^2}{\sigma_x^2}$.

of the variance of the explained variable, controlling for it will increase the variance of the estimator. Stated differently, controlling absorbs more of the variation in x than in the residual part of y ($y^{\perp x}$), it will increase the variance of the estimator.

Note that since in general for an outcome variable y, $R^2 = \text{corr}\,(\hat{y}, y)^2$ and since both regressions considered in the two R^2 above are univariate, we can rewrite the previous equivalence as

$$\sigma^2_{\text{CTRL}} > \sigma^2_{\text{OVB}} \quad \Leftrightarrow \quad \text{corr}(w, y^{\perp x})^2 < \text{corr}(w, x)^2$$

If the omitted variable is more correlated with x than with the residual part of y ($y^{\perp x}$), controlling for it will increase the variance of the estimator of interest.

Unessentially, note that we could rewrite the variance of the two estimators using the R^2 notations:

$$\sigma^2_{\text{OVB}} = \frac{\sigma^2_y(1 - R^2_{y\sim x})}{n\ \sigma^2_x} \qquad \text{and} \qquad \sigma^2_{\text{CTRL}} = \frac{\sigma^2_y(1 - R^2_{y\sim x,w})}{n\ \sigma^2_x(1 - R^2_{x\sim w})}$$

III. 1.2 ... and exaggeration

Now that we have shown that the variance of the estimator can be greater when we control than when we do not, we will see that exaggeration can be greater when we control than when we do not. The following discussion is mostly illustrative. A formal proof is derived in the following section.

Under the usual assumptions leading estimators to be normally distributed, we can apply corollary 1 stating that exaggeration of a normally distributed estimator increases with the variance of the estimator. and noting that $b_{\text{CTRL}} = 0$, we can rewrite the exaggeration ratio in the case where we control for the omitted variable described in equation 5 as:

$$E_{\text{CTRL}} = 1 + \frac{\sigma_{\text{CTRL}}}{\beta_1}\lambda\left(\frac{\beta_1}{\sigma_{\text{CTRL}}} - z_\alpha\right)$$

If controlling for w absorbs most of the variation in x without absorbing much of the variation in $y^{\perp x}$ (*i.e.* $\text{Var}(x^{\perp w}) \ll \text{Var}(x)$ while $\text{Var}(y^{\perp x,w}) < \text{Var}(y^{\perp x})$), σ_{CTRL} can be arbitrarily large. Thus, we can compute the limit of E_{CTRL} when $\sigma_{\text{CTRL}} \to \infty$:

$$\lim_{\sigma_{\text{CTRL}}\to\infty} E_{\text{CTRL}} = \lim_{\sigma_{\text{CTRL}}\to\infty} 1 + \frac{\sigma_{\text{CTRL}}}{\beta_1}\lambda\left(-z_\alpha\right) = +\infty$$

Controlling for a variable that absorbs most of the variation in x can cause exaggeration to go to infinity.

for any fixed amount of bias, we can have arbitrarily large standard errors for the estimator, depending on the parameter values, essentially, depending on how . We do that in section but skip the technicalities here.

Under the usual assumptions leading estimators to be normally distributed, we can combine the result above and corollary 1 stating that exaggeration of a normally distributed estimator increases when the variance of the estimator increase. Under these usual assumptions,

IV Asymptotic distribution of the estimators

As discussed in section I. 1, the goal is to compare the exaggeration ratios in the case of the naive regression to that of the corresponding reduced form and IV. As discussed in section II, in order to compute the exaggeration ratios for the various estimators, we need to know their distributions. In this section, we thus **derive the asymptotic distributions of the various estimators of interest**.

IV. 1 Naive regression (OVB)

Lemma 2. *Based on the data generating process described in equation 1, for the naive OLS estimate $\hat{\beta}_{\text{OVB}}$ of the regression of y on x, we have:*

$$\sqrt{n}\left(\hat{\beta}_{\text{OVB}} - \left(\beta_1 + \frac{\delta\gamma\sigma_w^2}{\sigma_x^2}\right)\right) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}{\sigma_x^2}\right)$$

Proof. We first compute the bias of the estimator. From equation 7 we get:

$$\begin{aligned}
& X'y = X'X\boldsymbol{\beta}_{\text{OVB}} + X'v \\
\Rightarrow\ & \mathbb{E}[X'y] = \underbrace{\mathbb{E}[X'X]}_{\text{pos. def.}}\boldsymbol{\beta}_{\text{OVB}} + \underbrace{\mathbb{E}[X'v]}_{0} \\
\Leftrightarrow\ & \boldsymbol{\beta}_{\text{OVB}} = \mathbb{E}[X'X]^{-1}\mathbb{E}[X'y] \\
\Leftrightarrow\ & \boldsymbol{\beta}_{\text{OVB}} = \mathbb{E}[X'X]^{-1}\mathbb{E}[X'(X\boldsymbol{\beta} + \delta w + u)] \qquad \text{cf eq. 1} \\
\Leftrightarrow\ & \boldsymbol{\beta}_{\text{OVB}} = \boldsymbol{\beta} + \mathbb{E}[X'X]^{-1}\mathbb{E}[X'w]\delta + \underbrace{\mathbb{E}[X'u]}_{0} \\
\Leftrightarrow\ & \boldsymbol{\beta}_{\text{OVB}} = \boldsymbol{\beta} + \mathbb{E}[X'X]^{-1}\mathbb{E}[X'w]\delta
\end{aligned} \tag{9}$$

Since X and w are correlated and $\delta \neq 0$, $\boxed{\boldsymbol{\beta}_{\text{OVB}} \neq \boldsymbol{\beta}}$. $\mathbb{E}[X'X]^{-1}\mathbb{E}[X'w]$ is the coefficient matrix of the projection of w on X. A similar derivation and discussion is carried out in Hansen (2022) for instance.

Let $\hat{\boldsymbol{\beta}}_{\text{OVB}}$ denote the least square estimator of $\boldsymbol{\beta}_{\text{OVB}}$. We thus have:

$$\hat{\boldsymbol{\beta}}_{\text{OVB}} = (X'X)^{-1}X'y \tag{10}$$

$$\begin{aligned}
\Rightarrow\ & \hat{\boldsymbol{\beta}}_{\text{OVB}} = (X'X)^{-1}X'(X\boldsymbol{\beta}_{\text{OVB}} + v) && \text{cf eq. 7} \\
\Rightarrow\ & \mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{OVB}} \mid X] = \boldsymbol{\beta}_{\text{OVB}} + (X'X)^{-1}\underbrace{\mathbb{E}[X'v]}_{0} \\
\Rightarrow\ & \mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{OVB}}] = \mathbb{E}[\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{OVB}} \mid X]] = \boldsymbol{\beta}_{\text{OVB}} && \text{by the Law of Iterated Expect. (LIE)}
\end{aligned}$$

Thus, <u>$\hat{\boldsymbol{\beta}}_{\text{OVB}}$ is an unbiased estimator of $\boldsymbol{\beta}_{\text{OVB}}$ but a biased estimator of $\boldsymbol{\beta}$</u>. We then compute its asymptotic distribution.

We can write:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{OVB}} - \boldsymbol{\beta}_{\text{OVB}}) = \left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i\mathrm{x}_i'\right)^{-1} \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i v_i\right)$$

By the Weak Law of Large Numbers (WLLN), $\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i\mathrm{x}_i' \xrightarrow{p} \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']$.
Since $\mathbb{E}[\mathrm{x}_i v_i] = 0$, by the Central Limit Theorem (CLT), $\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i v_i\right) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2])$.
Applying Slutsky's theorem then yields:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{OVB}} - \boldsymbol{\beta}_{\text{OVB}}) \xrightarrow{d} \mathcal{N}\left(0, \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2]\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\right)$$

And thus,

$$\boxed{\hat{\boldsymbol{\beta}}_{\text{OVB}} \xrightarrow{d} \mathcal{N}\left(\boldsymbol{\beta} + \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i w_i]\delta, \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2]\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}/n\right)} \tag{11}$$

We are interested in the second component of $\hat{\boldsymbol{\beta}}_{\text{OVB}}$. To retrieve it we need to compute $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}$, $\mathbb{E}[\mathrm{x}_i w_i]$ and $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2]$.

$$\mathbb{E}[\mathrm{x}_i\mathrm{x}_i'] = \mathbb{E}\begin{bmatrix} 1 & x \\ x & x^2 \end{bmatrix} = \begin{bmatrix} 1 & \mu_x \\ \mu_x & \sigma_x^2 + \mu_x^2 \end{bmatrix} \quad \Rightarrow \quad \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1} = \frac{1}{\sigma_x^2}\begin{bmatrix} \sigma_x^2 + \mu_x^2 & -\mu_x \\ -\mu_x & 1 \end{bmatrix}$$

$$\mathbb{E}[\mathrm{x}_i w_i] = \mathbb{E}\begin{bmatrix} w_i \\ x_i w_i \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbb{E}[x_i]\underbrace{\mathbb{E}[w_i]}_{0} + \mathrm{cov}(x_i, w_i) \end{bmatrix} = \begin{bmatrix} 0 \\ \pi_1 \underbrace{\mathrm{cov}(z_i, w_i)}_{0} + \gamma \underbrace{\mathrm{var}(w_i)}_{\sigma_w^2} + \underbrace{\mathrm{cov}(\epsilon_i, w_i)}_{0} \end{bmatrix} = \begin{bmatrix} 0 \\ \gamma\sigma_w^2 \end{bmatrix}$$

$$\Rightarrow \quad \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i w_i] = \frac{\gamma\sigma_w^2}{\sigma_x^2}\begin{bmatrix} -\mu_x \\ 1 \end{bmatrix} \tag{12}$$

Note that $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2] \overset{\text{LIE}}{=} \mathbb{E}[\mathrm{x}_i\mathrm{x}_i'\mathbb{E}[v_i^2|\mathrm{x}_i]]$. We thus first compute $\mathbb{E}[v_i^2|\mathrm{x}_i]$, noting that:

$$\begin{aligned}
v_i &= y_i - \mathrm{x}_i'\boldsymbol{\beta}_{\text{OVB}} \\
&= \delta w_i + u_i + \mathrm{x}_i'(\boldsymbol{\beta} - \boldsymbol{\beta}_{\text{OVB}}) \\
&= \delta w_i + u_i - \underbrace{\mathrm{x}_i'\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i w_i]}_{\text{projection of } w_i \text{ on } x_i}\delta \\
&= u_i + \delta\underbrace{\left(w_i - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)\right)}_{\text{part of } w_i \text{ orthogonal to } x_i} \\
&= u_i + \delta w_i^{\perp} \qquad\qquad \text{where } w_i^{\perp} = w_i - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)
\end{aligned}$$

And thus,

$$\begin{aligned}
\mathbb{E}[v_i^2|\mathrm{x}_i] &= \mathbb{E}[(u_i + \delta w_i^{\perp})^2|x_i] \\
&= \mathbb{E}[u_i^2|x_i] + 2\delta\mathbb{E}[u_i w_i^{\perp}|x_i] + \delta^2\mathbb{E}[(w_i^{\perp})^2|x_i] \\
&= \sigma_u^2 + 2\delta\left(\mathbb{E}[u_i w_i|x_i] - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)\underbrace{\mathbb{E}[u_i|x_i]}_{0}\right) + \delta^2\mathbb{E}[(w_i^{\perp})^2|x_i] \\
&\overset{\text{LIE}}{=} \sigma_u^2 + 2\delta\mathbb{E}[w_i\underbrace{\mathbb{E}[u_i|x_i, w_i]}_{0}|x_i] + \delta^2\mathbb{E}[(w_i^{\perp})^2|x_i] \\
&= \sigma_u^2 + \delta^2\mathbb{E}[(w_i^{\perp})^2|x_i]
\end{aligned}$$

Notice that, by the law of total variance, $\mathbb{E}[(w_i^{\perp})^2|x_i] = \mathrm{Var}(w_i^{\perp}|x_i) + \mathbb{E}[w_i^{\perp}|x_i]^2$. Now, since $w_i^{\perp}$ is the component of w_i that is orthogonal to x_i and by the projection interpretation of the conditional variance, $\mathbb{E}[w_i^{\perp}|x_i] = 0$. And thus, since by assumption $\mathrm{Var}(w_i^{\perp}|x_i) = \mathrm{Var}(w_i^{\perp})$,

$$\begin{aligned}
\mathbb{E}[(w_i^{\perp})^2|x_i] &= \mathrm{Var}(w_i^{\perp}|x_i) \\
&= \mathrm{Var}(w_i^{\perp}) \\
&= \mathbb{E}[(w_i^{\perp})^2] - \mathbb{E}[w_i^{\perp}]^2 \\
&= \mathbb{E}\left[\left(w_i - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)\right)^2\right] - \left(\underbrace{\mathbb{E}[w_i]}_{0} + \frac{\gamma\sigma_w^2}{\sigma_x^2}\underbrace{\mathbb{E}[x_i - \mu_x]}_{0}\right)^2 \\
&= \underbrace{\mathbb{E}[w_i^2]}_{\sigma_w^2} - 2\frac{\gamma\sigma_w^2}{\sigma_x^2}\left(\underbrace{\mathbb{E}[x_i w_i]}_{\gamma\sigma_w^2} - \mu_x\underbrace{\mathbb{E}[w_i]}_{0}\right) + \frac{\gamma^2\sigma_w^4}{\sigma_x^4}\underbrace{\mathbb{E}[(x_i - \mu_x)^2]}_{\sigma_x^2} \\
&= \sigma_w^2 - 2\frac{\gamma^2\sigma_w^4}{\sigma_x^2} + \frac{\gamma^2\sigma_w^4}{\sigma_x^2} \\
&= \sigma_w^2 - \frac{\gamma^2\sigma_w^4}{\sigma_x^2} \\
&= \sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)
\end{aligned}$$

Note that this variance is well defined (positive) only if $\sigma_x^2 \geq \gamma^2\sigma_w^2$. Under this condition,

$$\mathbb{E}[v_i^2|\mathrm{x}_i] = \sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right) \tag{13}$$

Thus, under our set of assumptions, $\mathbb{E}[v_i^2|\mathrm{x}_i]$ does not depend on x_i and $\mathbb{E}[v_i^2|\mathrm{x}_i] = \mathbb{E}[v_i^2]$. We denote this quantity σ_v^2.

We can now compute the variance of the estimator $\hat{\boldsymbol{\beta}}_{\mathrm{OVB}}$, noting that $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2] = \mathbb{E}[\mathrm{x}_i\mathrm{x}_i'\mathbb{E}[v_i^2|\mathrm{x}_i]] = \mathbb{E}[\mathrm{x}_i\mathrm{x}_i'\sigma_v^2] = \sigma_v^2\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']$. And thus $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2]\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1} = \sigma_v^2\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']$.

Plugin this and equation 12 into equation 11, we get, for $\hat{\beta}_{\text{OVB}}$, the second component of $\hat{\boldsymbol{\beta}}_{\text{OVB}}$:

$$\hat{\beta}_{\text{OVB}} \xrightarrow{d} \mathcal{N}\left(\beta_1 + \frac{\delta\gamma\sigma_w^2}{\sigma_x^2},\ \frac{\sigma_v^2}{n\ \sigma_x^2}\right)$$

And substituting for σ_v^2,

$$\boxed{\hat{\beta}_{\text{OVB}} \xrightarrow{d} \mathcal{N}\left(\beta_1 + \frac{\delta\gamma\sigma_w^2}{\sigma_x^2},\ \frac{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}{n\ \sigma_x^2}\right)}$$

□

Using the notations introduced in section I. 1, we have:

$$\begin{aligned} b_{\text{OVB}} &= \frac{\delta\gamma\sigma_w^2}{\sigma_x^2} \\ \sigma_{\text{OVB}}^2 &= \frac{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}{n\ \sigma_x^2} \end{aligned} \tag{14}$$

Note, in the absence of omitted variable bias, we retrieve the usual asymptotic distribution of the OLS estimator. To do so, we can:

— Set $\delta = 0$. In this case $b = 0$ and $\sigma_{\text{OVB}}^2 = \dfrac{\sigma_u^2}{n\sigma_x^2}$, the usual variance.

— Set $\sigma_w = 0$ yields the same bias and variance.

— Set $\gamma = 0$. In this case $b = 0$ and $\sigma_{\text{OVB}}^2 = \dfrac{\sigma_u^2 + \delta^2\sigma_w^2}{n\sigma_x^2}$. The numerator of this variance differs because the variance of y not explained by x includes the effect of w. It is therefore equivalent to the usual variance.

IV. 2 Instrumental Variable

Lemma 3. *Based on the sus-mentioned data generating process, for the IV estimate $\hat{\beta}_{\text{IV}}$ of the regression of y on x instrumented by z, we have:*

$$\sqrt{n}\left(\hat{\beta}_{\text{IV}} - \beta_1\right) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{\delta^2\sigma_w^2 + \sigma_u^2}{\pi_1^2\sigma_z^2}\right)$$

Proof. We know that we have:

$$\begin{aligned} \hat{\boldsymbol{\beta}}_{\text{IV}} &= (Z'X)^{-1}Z'y \\ &= (Z'X)^{-1}Z'(X\boldsymbol{\beta} + \delta w + u) \\ &= \boldsymbol{\beta} + (Z'X)^{-1}Z'w\delta + (Z'X)^{-1}Z'u \end{aligned}$$

Thus,

$$\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{IV}}|X, Z] = \boldsymbol{\beta} + (Z'X)^{-1}\underbrace{\mathbb{E}[Z'w]}_{0}\delta + (Z'X)^{-1}\underbrace{\mathbb{E}[Z'u]}_{0} = \boldsymbol{\beta} \quad \Rightarrow \quad \mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{IV}}] = \boldsymbol{\beta}$$

The IV estimator is unbiased. We now want to derive its asymptotic distribution.

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{IV}} - \boldsymbol{\beta}) = \left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{z}_i \mathrm{x}_i'\right)^{-1} \delta\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{z}_i w_i\right) + \left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{z}_i \mathrm{x}_i'\right)^{-1} \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{z}_i u_i\right)$$

The WLLN gives $\frac{1}{n}\sum_{i=1}^{n} \mathrm{z}_i \mathrm{x}_i' \xrightarrow{p} \mathbb{E}[\mathrm{z}_i \mathrm{x}_i']$ and since $\mathbb{E}[\mathrm{z}_i w_i] = 0$ and $\mathbb{E}[\mathrm{z}_i u_i] = 0$, the Central Limit Theorem (CLT) gives $\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{z}_i w_i\right) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{z}_i \mathrm{z}_i' w_i^2])$ and $\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{z}_i u_i\right) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{z}_i \mathrm{z}_i' u_i^2])$.

Now, $\mathbb{E}[\mathrm{z}_i \mathrm{z}_i' u_i^2] = \mathbb{E}[\mathrm{z}_i \mathrm{z}_i' \mathbb{E}[u_i^2|\mathrm{z}_i]] = \mathbb{E}[\mathrm{z}_i \mathrm{z}_i' \mathbb{E}(\mathbb{E}[u_i^2|\mathrm{x}_i]|\mathrm{z}_i)] = \mathbb{E}[\mathrm{z}_i \mathrm{z}_i' \mathbb{E}[\sigma_u^2|\mathrm{z}_i]] = \mathbb{E}[\mathrm{z}_i \mathrm{z}_i' \sigma_u^2] = \sigma_u^2 \mathbb{E}[\mathrm{z}_i \mathrm{z}_i']$. Now, since z_i is a valid instrument, it makes sense to have $\mathbb{E}[w_i^2|\mathrm{z}_i] = \sigma_w^2$. Thus, $\mathbb{E}[\mathrm{z}_i \mathrm{z}_i' w_i^2] = \mathbb{E}[\mathrm{z}_i \mathrm{z}_i' \mathbb{E}[w_i^2|\mathrm{z}_i]] = \sigma_w^2 \mathbb{E}[\mathrm{z}_i \mathrm{z}_i']$.

Applying Slutsky's theorem then yields:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{IV}} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}\left(0, (\delta^2\sigma_w^2 + \sigma_u^2)\mathbb{E}[\mathrm{z}_i \mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{z}_i \mathrm{z}_i']\left(\mathbb{E}[\mathrm{z}_i \mathrm{x}_i']^{-1}\right)'\right)$$

And thus,

$$\boxed{\hat{\boldsymbol{\beta}}_{\text{IV}} \xrightarrow{d} \mathcal{N}\left(\boldsymbol{\beta}, \frac{\delta^2\sigma_w^2 + \sigma_u^2}{n}\mathbb{E}[\mathrm{z}_i \mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{z}_i \mathrm{z}_i']\left(\mathbb{E}[\mathrm{z}_i \mathrm{x}_i']^{-1}\right)'\right)}$$

We are interested in the second component of $\hat{\boldsymbol{\beta}}_{\text{IV}}$. To retrieve it we need to compute $\mathbb{E}[\mathrm{z}_i \mathrm{z}_i]$, $\mathbb{E}[\mathrm{x}_i \mathrm{z}_i']^{-1}$ and its transpose.

$$\mathbb{E}[\mathrm{z}_i \mathrm{z}_i'] = \mathbb{E}\begin{bmatrix} 1 & z \\ z & z^2 \end{bmatrix} = \begin{bmatrix} 1 & \mu_z \\ \mu_z & \sigma_z^2 + \mu_z^2 \end{bmatrix}$$

$$\mathbb{E}[\mathrm{z}_i \mathrm{x}_i'] = \begin{bmatrix} 1 & \mathbb{E}[x_i] \\ \mathbb{E}[z_i] & \mathbb{E}[z_i x_i] \end{bmatrix} = \begin{bmatrix} 1 & \pi_0 + \pi_1\mathbb{E}[z_i] + \gamma\underbrace{\mathbb{E}[w_i]}_{0} + \underbrace{\mathbb{E}[\epsilon_i]}_{0} \\ \mu_z & \pi_0\mathbb{E}[z_i] + \pi_1\mathbb{E}[z_i^2] + \gamma\underbrace{\mathbb{E}[z_i w_i]}_{0} + \underbrace{\mathbb{E}[z_i \epsilon_i]}_{0} \end{bmatrix} = \begin{bmatrix} 1 & \pi_0 + \pi_1\mu_z \\ \mu_z & \pi_0\mu_z + \pi_1(\sigma_z^2 + \mu_z^2) \end{bmatrix}$$

$$\Rightarrow \quad \mathbb{E}[\mathrm{z}_i \mathrm{x}_i']^{-1} = \frac{1}{\pi_1\sigma_z^2}\begin{bmatrix} \pi_0\mu_z + \pi_1(\sigma_z^2 + \mu_z^2) & -\pi_0 - \pi_1\mu_z \\ -\mu_z & 1 \end{bmatrix}$$

$$\text{and} \quad \left(\mathbb{E}[\mathrm{z}_i \mathrm{x}_i']^{-1}\right)' = \frac{1}{\pi_1\sigma_z^2}\begin{bmatrix} \pi_0\mu_z + \pi_1(\sigma_z^2 + \mu_z^2) & -\mu_z \\ -\pi_0 - \pi_1\mu_z & 1 \end{bmatrix}$$

Thus,

$$\mathbb{E}[\mathrm{z}_i \mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{z}_i \mathrm{z}_i'] = \frac{1}{\pi_1\sigma_z^2}\begin{bmatrix} \pi_1\sigma_z^2 & -\pi_0\sigma_z^2 \\ 0 & \sigma_z^2 \end{bmatrix} = \begin{bmatrix} 1 & -\frac{\pi_0}{\pi_1} \\ 0 & \frac{1}{\pi} \end{bmatrix}$$

And finally,

$$\mathbb{E}[\mathrm{z}_i \mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{z}_i \mathrm{z}_i']\left(\mathbb{E}[\mathrm{z}_i \mathrm{x}_i']^{-1}\right)' = \frac{1}{\pi_1\sigma_z^2}\begin{bmatrix} 2\pi_0\mu_z + \pi_1(\sigma_z^2 + \mu_z^2) + \frac{\pi_0^2}{\pi_1} & -\mu_z - \frac{\pi_0}{\pi_1} \\ -\mu_z - \frac{\pi_0}{\pi_1} & \frac{1}{\pi_1} \end{bmatrix}$$

Thus, for $\hat{\beta}_{\text{IV}}$, the second component of $\hat{\boldsymbol{\beta}}_{\text{IV}}$, we have:

$$\boxed{\hat{\beta}_{\text{IV}} \xrightarrow{d} \mathcal{N}\left(\beta_1,\ \frac{\delta^2\sigma_w^2 + \sigma_u^2}{n\ \pi_1^2\sigma_z^2}\right)} \tag{15}$$

□

Using the notations introduced in section I. 1, we have:

$$\begin{aligned} b_{\text{IV}} &= 0 \\ \sigma^2_{\text{IV}} &= \frac{\delta^2\sigma_w^2 + \sigma_u^2}{n\ \pi_1^2\sigma_z^2} \end{aligned} \tag{16}$$

Note that, since z_i is uncorrelated with w_i and ϵ_i,

$$\begin{aligned} \sigma_x^2 &= \text{var}(x_i) = \text{var}(\pi_0 + \pi_1 z_i + \gamma w_i + \epsilon_i) \\ &= \pi_1^2\sigma_z^2 + \gamma^2\sigma_w^2 + \sigma_\epsilon^2 + 2\pi_1\gamma\underbrace{\text{cov}(z_i, w_i)}_{0} + 2\pi_1\underbrace{\text{cov}(z_i, \epsilon_i)}_{0} + 2\gamma\underbrace{\text{cov}(w_i, \epsilon_i)}_{0} \\ &= \underbrace{\pi_1^2\sigma_z^2}_{V_z^x} + \underbrace{\gamma^2\sigma_w^2 + \sigma_\epsilon^2}_{V^x_{\text{non-}z}} \end{aligned}$$

For a given variance of x, <u>the larger the proportion of this variance that depends on z, V_z^x, (and the smaller the non-z component $V^x_{\text{non-}z}$), the more precise the estimate</u>.

Also note that,

$$\begin{aligned} & \text{cov}(x_i, z_i) = \text{cov}(\pi_0 + \pi_1 z_i + \gamma w_i + \epsilon_i, z_i) = \pi_1\sigma_z^2 \\ \Rightarrow\quad & \rho_{xz} = \text{corr}(x_i, z_i) = \frac{\text{cov}(x_i, z_i)}{\sigma_x\sigma_z} = \pi_1\frac{\sigma_z}{\sigma_x} \\ \Rightarrow\quad & \pi_1^2\sigma_z^2 = \rho_{xz}^2\sigma_x^2 \end{aligned}$$

One can therefore rewrite the formula for the variance of our IV estimator as

$$\sigma^2_{\text{IV}} = \frac{\delta^2\sigma_w^2 + \sigma_u^2}{n\ \sigma_x^2\rho_{xz}^2}$$

This variance formula enables to retrieve the usual result regarding the determinant of the 2SLS variance as discussed in Hansen (2022) for instance. The variance of the estimator increases when σ_u^2, the variance of the error, increases. It decreases with ρ_{xz}, the correlation between x and z, and with σ_x^2, the variance of x, assuming the omitted variable fixed.

IV. 3 Reduced form

Lemma 4. *Based on the sus-mentioned data generating process, for the reduced form estimate*

$\hat{\beta}_{\text{RED}}$ of the regression of y on d, we have:

$$\sqrt{n}\left(\hat{\beta}_{\text{RED}} - \beta_1\right) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{(\delta+\beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_\epsilon^2}{\sigma_d^2}\right)$$

with $\sigma_d^2 = p_1(1-p_1)$, p_1 being the proportion of treated observations, i.e., the proportion of 1 in d.

Proof. We first derive the bias of the estimator.

$$\begin{aligned}
\hat{\boldsymbol{\beta}}_{\text{RED}} &= (D'D)^{-1}D'y \\
&= (D'D)^{-1}D'(D\boldsymbol{\beta} + \kappa w + \nu) \\
&= \boldsymbol{\beta} + \kappa(D'D)^{-1}D'w + (D'D)^{-1}D'\nu
\end{aligned}$$

Thus,

$$\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{RED}}|D] = \boldsymbol{\beta} + (D'D)^{-1}\underbrace{\mathbb{E}[D'w]}_{0}\kappa + (D'D)^{-1}\underbrace{\mathbb{E}[D'\nu]}_{\substack{\beta_1\mathbb{E}[D'\epsilon]+\mathbb{E}[D'u] \\ = \beta_1 0 + 0 \,=\, 0}} = \boldsymbol{\beta}$$

By the Law of Iterated Expectations, we get that $\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{RED}}] = \mathbb{E}[\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{RED}}|D]] = \beta$ and thus $\hat{\boldsymbol{\beta}}_{\text{RED}}$ is an unbiased estimator of $\boldsymbol{\beta}$. We then derive its asymptotic distribution, noting that:

$$\begin{aligned}
\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{RED}} - \boldsymbol{\beta}) &= \left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i\mathrm{d}_i'\right)^{-1}\kappa\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i w_i\right) + \left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i\mathrm{d}_i'\right)^{-1}\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i\nu_i\right) \\
&= \left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i\mathrm{d}_i'\right)^{-1}\kappa\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i w_i\right) \\
&\quad + \left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i\mathrm{d}_i'\right)^{-1}\left[\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i u_i\right) + \beta_1\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i\epsilon_i\right)\right]
\end{aligned}$$

The WLLN gives $\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i\mathrm{d}_i' \xrightarrow{p} \mathbb{E}[\mathrm{d}_i\mathrm{d}_i']$ and since $\mathbb{E}[\mathrm{d}_i w_i] = 0$,$\mathbb{E}[\mathrm{d}_i u_i] = 0$ and $\mathbb{E}[\mathrm{d}_i\epsilon_i] = 0$, the Central Limit Theorem (CLT) gives:

$$\begin{aligned}
\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i w_i\right) &\xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{d}_i\mathrm{d}_i' w_i^2]) \\
\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i u_i\right) &\xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{d}_i\mathrm{d}_i' u_i^2]) \\
\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{d}_i \epsilon_i\right) &\xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{d}_i\mathrm{d}_i' \epsilon_i^2])
\end{aligned}$$

As for the IV, $\mathbb{E}[\mathrm{d}_i\mathrm{d}_i' u_i^2] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\mathbb{E}[u_i^2|\mathrm{d}_i]] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\mathbb{E}(\mathbb{E}[u_i^2|\mathrm{x}_i]|\mathrm{d}_i)] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\mathbb{E}[\sigma_u^2|\mathrm{d}_i]] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\sigma_u^2] = \sigma_u^2\mathbb{E}[\mathrm{d}_i\mathrm{d}_i']$. In addition, $\mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\epsilon_i^2] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\mathbb{E}[\epsilon_i^2|\mathrm{d}_i]] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\sigma_\epsilon^2] = \sigma_\epsilon^2\mathbb{E}[\mathrm{d}_i\mathrm{d}_i']$. Now, since d_i is exogenous, it makes sense to have $\mathbb{E}[w_i^2|\mathrm{d}_i] = \sigma_w^2$. Thus, $\mathbb{E}[\mathrm{d}_i\mathrm{d}_i' w_i^2] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\mathbb{E}[w_i^2|\mathrm{d}_i]] = \sigma_w^2\mathbb{E}[\mathrm{d}_i\mathrm{d}_i']$.

Applying Slutsky's theorem then yields:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{RED}} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}\left(0, (\kappa^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_\epsilon^2)\mathbb{E}[\mathrm{d}_i\mathrm{d}_i']^{-1}\right)$$

Now, since $\kappa = \delta + \beta\gamma$,

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{RED}} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}\left(0, \{(\delta + \beta\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_\epsilon^2\}\mathbb{E}[\mathrm{d}_i\mathrm{d}_i']^{-1}\right)$$

We are interested in the second component of $\hat{\boldsymbol{\beta}}_{\text{RED}}$. To retrieve it we need to compute $\mathbb{E}[\mathrm{d}_i\mathrm{d}_i']^{-1}$.

$$\mathbb{E}[\mathrm{d}_i\mathrm{d}_i'] = \mathbb{E}\begin{bmatrix} 1 & d \\ d & d^2 \end{bmatrix} = \begin{bmatrix} 1 & \mu_d \\ \mu_d & \sigma_d^2 + \mu_d^2 \end{bmatrix} \quad \Rightarrow \quad \mathbb{E}[\mathrm{d}_i\mathrm{d}_i']^{-1} = \frac{1}{\sigma_d^2}\begin{bmatrix} \sigma_d^2 + \mu_d^2 & -\mu_d \\ -\mu_d & 1 \end{bmatrix}$$

Thus, for $\hat{\beta}_{\text{RED}}$, the second component of $\hat{\boldsymbol{\beta}}_{\text{RED}}$, we have:

$$\boxed{\hat{\boldsymbol{\beta}}_{\text{RED}} \xrightarrow{d} \mathcal{N}\left(\boldsymbol{\beta}, \frac{(\delta + \beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_\epsilon^2}{n\sigma_d^2}\right)}$$

□

Using the notations introduced in section I. 1, we have:

$$\begin{aligned} b_{\text{RED}} &= 0 \\ \sigma^2_{\text{RED}} &= \frac{(\delta + \beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_\epsilon^2}{n\sigma_d^2} \end{aligned} \tag{17}$$

Notice that, as expected, the numerator of the variance corresponds to the part of the variance in y that is orthogonal to d, *i.e.*, the variance of the error term of the regression of y on d:

$$\begin{aligned} \mathrm{Var}\left(y_i^{\perp d}\right) &= \mathrm{Var}\left((\delta + \beta_1\gamma)w_i + u_i + \beta_1\epsilon_i\right) \\ &= (\delta + \beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_\epsilon^2 \\ &\quad + 2(\delta + \beta_1\gamma)\underbrace{\mathrm{cov}(w_i, u_i)}_{0} + 2\beta_1\underbrace{\mathrm{cov}(u_i, \epsilon_i)}_{0} + 2(\delta + \beta_1\gamma)\beta_1\underbrace{\mathrm{cov}(w_i, \epsilon_i)}_{0} \\ &= (\delta + \beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_\epsilon^2 \end{aligned}$$

Further, since d is a vector of dummies, denoting p_1 the proportion of 1 in this vector, $\sigma_d^2 = p_1(1-p_1)$ and thus:

$$\hat{\boldsymbol{\beta}}_{\text{RED}} \xrightarrow{d} \mathcal{N}\left(\boldsymbol{\beta}, \frac{(\delta + \beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_\epsilon^2}{n \times p_1(1 - p_1)}\right) \tag{18}$$

For all other parameters fixed, including the number of observations n and the true effect size β_1, σ^2_{RED} increases when the proportion of treated observations p_1 gets further from 0.5 as $p_1(1 - p_1)$ is a parabolic function with maximum reached for $p_1 = 0.5$. When most observations have the same treatment status, *i.e.*, p_1 close to 0 or 1, the variance shoots up. There is not enough variation in the treatment status to precisely identify the effect of interest.

IV. 4 Comparison of efficiencies

The IV and naive regression variances are rather similar. Note that $\sigma_x^2\rho_{xz}^2 < \sigma_x^2$ and,

$$\frac{\gamma^2\sigma_w^2}{\sigma_x^2} > 0 \quad \Leftrightarrow \quad 1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2} < 1 \quad \Leftrightarrow \quad \sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right) < \sigma_u^2 + \delta^2\sigma_w^2$$

Both these inequalities imply that $\boxed{\sigma_{\text{IV}}^2 > \sigma_{\text{OVB}}^2}$. While unbiased, the IV estimator is less precise than the naive estimator. This is a well known and intuitive result that we recovered in a particular setting, tied to the modeling assumptions we made. This result is due to the fact that the IV only uses part of the variation in the endogenous dependent variable, the part that is explained by the instrument.

Focusing on the setting considered for the reduced form estimation, *i.e.*, with $\pi_0 = 0,\ \pi_1 = 1$ and z denoted with d, the asymptotic distribution of the IV estimate in this setting would be:

$$\sqrt{n}\left(\hat{\beta}_{\text{IV(RED)}} - \beta_1\right) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{\delta^2\sigma_w^2 + \sigma_u^2}{\sigma_d^2}\right)$$

We can notice that $\sigma_{\text{RED}}^2 = \sigma_{\text{IV(RED)}}^2 + \dfrac{2\beta_1\gamma\sigma_w^2 + \beta_1^2\gamma^2 + \beta_1^2\sigma_\epsilon^2}{\sigma_d^2}$. We assumed $\beta_1 > 0$. It makes sense to focus on the case for which $\gamma > 0$ and $\delta > 0$, *i.e.*, the OVB described in equation 14 is an upward bias and causes the same types of issues as the exaggeration. In this case $\sigma_{\text{RED}}^2 > \sigma_{\text{IV(RED)}}^2$. In the special setting considered for the reduced form estimation and under these assumptions, we thus have $\boxed{\sigma_{\text{RED}}^2 > \sigma_{\text{IV(RED)}}^2 > \sigma_{\text{OVB(RED)}}^2}$.

V Comparison of exaggeration ratios

Now that we have computed the asymptotic distributions of the various estimators, we can compute their exaggeration ratios. We study how they evolve with the values of model parameters and prove the existence of the exaggeration/confounder trade-off in our specific case.

V. 1 Evolution of the exaggeration ratio with the value of model parameters

In this section, we aim to understand how the exaggeration ratios of the naive regression, the IV and the reduced form evolves with δ and γ, the intensity of the OVB, ρ_{xz}, the correlation between x and z, and p_1 the proportion of treated observations, respectively.

V. 1.1 Naive regression

Plugging in the values for β_{OVB}, σ_{OVB} and b_{OVB}, from equation 14 into equation 5, we get:

$$E_{\text{OVB}} = 1 + \frac{\delta\gamma\sigma_w^2}{\beta_1\sigma_x^2} + \frac{\sqrt{\dfrac{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}{n\ \sigma_x^2}}}{\beta_1}\lambda\left(\frac{\beta_1 + \dfrac{\delta\gamma\sigma_w^2}{\sigma_x^2}}{\sqrt{\dfrac{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}{n\ \sigma_x^2}}} - z_\alpha\right)$$

$$= 1 + \frac{\delta\gamma\sigma_w^2}{\beta_1\sigma_x^2} + \frac{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}}{\beta_1\sigma_x\sqrt{n}}\lambda\left(\frac{\sigma_x\sqrt{n}\left(\beta_1 + \dfrac{\delta\gamma\sigma_w^2}{\sigma_x^2}\right)}{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}} - z_\alpha\right)$$

Keeping in mind that λ is a decreasing function, we notice that E_{ovb} decreases with β_1, σ_x and n and increases with σ_u. Analytically deriving the evolution of this quantity with the omitted variable bias, *ie*, with δ, γ and σ_w is more challenging. The direction of this evolution is likely to depend on the value of the parameters. Since this is not the focus of the present paper, we do not dive into this. We however expect that, if the bias is upward (δ and γ of the same sign), we expect the exaggeration to increase with δ, γ and σ_w^2, for realistic values of the various parameters since the usual omitted variable bias $\left(\frac{\delta\gamma\sigma_w^2}{\sigma_x^2}\right)$ is a major component of the exaggeration.

V. 1.2 Instrumental Variable

Plugging in the values for β_{IV}, σ_{IV} and b_{IV}, from equation 16 into equation 5, we get:

$$E_{\text{IV}} = 1 + \frac{\sqrt{\dfrac{\sigma_u^2 + \delta^2\sigma_w^2}{n\ \pi_1^2\sigma_z^2}}}{\beta_1}\lambda\left(\frac{\beta_1}{\sqrt{\dfrac{\sigma_u^2 + \delta^2\sigma_w^2}{n\ \pi_1^2\sigma_z^2}}} - z_\alpha\right)$$

$$= 1 + \frac{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2}}{\beta_1|\pi_1|\sigma_z\sqrt{n}}\lambda\left(\frac{\beta_1|\pi_1|\sigma_z\sqrt{n}}{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2}} - z_\alpha\right)$$

Proposition 1. *E_{IV}, the exaggeration ratio of the instrumental variable estimator, decreases with the strength of the instrument π_1 (or equivalently with the correlation between the endegoneous variable x and the instrument z).*

Proof. Since there is no bias, the only factor varying with the values of the model parameters is the standard error of the estimator. Corollary 1 showed the exaggeration ratio increases with this standard error. Thus, any model parameter positively associated with the later will increase the former. Hence, the exaggeration ratio of the IV estimator, E_{IV}, increases with σ_u, σ_w and δ and decreases with σ_z, π_1 and n. It also decreases with β_1. We thus showed that exaggeration decreases with the strength of the instrument. □

Now, note that:

$$\begin{aligned}
\sigma_y^2 &= \text{var}(y_i) = \text{var}(\beta_0 + \beta_1 x_i + \delta w_i + u_i) \\
&= \beta_1^2\sigma_x^2 + \delta^2\sigma_w^2 + \sigma_u^2 + 2\beta_1\delta\underbrace{\text{cov}(x_i, w_i)}_{\gamma\sigma_w^2} + 2\beta_1\underbrace{\text{cov}(x_i, u_i)}_{0} + 2\delta\underbrace{\text{cov}(w_i, u_i)}_{0} \\
&= \beta_1^2\sigma_x^2 + (\delta^2 + 2\beta_1\gamma\delta)\sigma_w^2 + \sigma_u^2 \\
&= \beta_1^2(\pi_1^2\sigma_z^2 + \gamma^2\sigma_w^2 + \sigma_\epsilon^2) + (\delta^2 + 2\beta_1\gamma\delta)\sigma_w^2 + \sigma_u^2 \\
&= \underbrace{\beta_1^2\pi_1^2\sigma_z^2}_{V_z^y} + \underbrace{(\beta_1\gamma + \delta)^2\sigma_w^2 + (\beta_1^2\sigma_\epsilon^2 + \sigma_u^2)}_{V_{\text{non-}z}^y}
\end{aligned}$$

The larger the parameters that are positively associated with E_{IV} (σ_u, σ_w and δ), the larger the proportion of the variance of y that does not depend on z, $V_{\text{non-}z}^y$. Hence, for a given variance of y, the smaller the proportion of this variance that depends on z, V_z^y, (and the larger the non-z component $V_{\text{non-}z}^y$), the less precise the estimate and the larger the exaggeration. The smaller the proportion of the variance in y that depends on z, the larger the exaggeration of the IV estimator.

Since $\pi_1^2\sigma_z^2 = \rho_{xz}^2\sigma_x^2$, we can also express this exaggeration ratio as:

$$E_{\text{IV}} = 1 + \frac{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2}}{\beta_1\rho_{xz}\sigma_x\sqrt{n}}\lambda\left(\frac{\beta_1\rho_{xz}\sigma_x\sqrt{n}}{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2}} - z_\alpha\right)$$

Note that E_{IV} decreases with ρ_{xz}, the correlation between the endogenous variable x and the instrument z.

V. 1.3 Reduced form

Plugging in the values for β_{RE}, σ_{RE} and b_{RED}, from equation 17 into equation 5, we get:

$$E_{\text{RED}} = 1 + \frac{\sqrt{(\delta + \beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_\epsilon^2}}{\beta_1\sqrt{n \times p_1(1-p_1)}}\lambda\left(\frac{\beta_1\sqrt{n \times p_1(1-p_1)}}{\sqrt{(\delta + \beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_\epsilon^2}} - z_\alpha\right)$$

Proposition 2. *E_{RED}, the exaggeration ratio of the reduced form estimator, increases when the proportion of treated observations moves away from 0.5.*

Proof. Equation 18 showed that σ_{RED}^2 increases when the proportion of treated observations moves away from 0.5. Applying corollary 1 directly proves the proposition. □

V. 2 The exaggeration/confounders trade-off

The objective of this paper is to compare "biases" resulting from exaggeration and confounders. In this document, we thus aim to compare E_{OVB} to E_{IV} and E_{RED}. We thus define $\Delta_{\text{IV}} = E_{\text{IV}} - E_{\text{OVB}}$ and $\Delta_{\text{RED}} = E_{\text{RED}} - E_{\text{OVB}}$ and analyse whether and for which parameters values these quantities are positive. In particular, we focus on the strength of the IV (or the correlation between the instrument and the endogenous variable) and the proportion of treated observations in the reduced form case.

V. 2.1 Instrumental variable

Proposition 3. *Under the data generating process described in equation 1, for some parameter values, $E_{\text{IV}} > E_{\text{OVB}}$. More specifically, $\exists \pi_1 \in \mathbb{R}_+^*$ s.t. $E_{\text{IV}} > E_{\text{OVB}}$.*

Proof. Combining lemmas 2, 3 and 1, we get:

$$\Delta_{\text{IV}} = E_{\text{IV}} - E_{\text{OVB}} = \frac{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2}}{\beta_1\rho_{xz}\sigma_x\sqrt{n}}\lambda\left(\frac{\beta_1\rho_{xz}\sigma_x\sqrt{n}}{\sqrt{\sigma_u^2+\delta^2\sigma_w^2}} - z_\alpha\right) - \frac{\delta\gamma\sigma_w^2}{\beta_1\sigma_x^2}$$
$$- \frac{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2\left(1-\frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}}{\beta_1\sigma_x\sqrt{n}}\lambda\left(\frac{\sigma_x\sqrt{n}\left(\beta_1 + \frac{\delta\gamma\sigma_w^2}{\sigma_x^2}\right)}{\sqrt{\sigma_u^2+\delta^2\sigma_w^2\left(1-\frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}} - z_\alpha\right)$$

We want to show that $\exists \pi_1 \in \mathbb{R}_+^*$ s.t. $\Delta_{\text{IV}} > 0$. First, note that only E_{IV} depends on ρ_{xz}. Since E_{IV} decreases with ρ_{xz}, Δ_{IV} also decreases with ρ_{xz}. Now, for fixed and finite values of the other parameters, and since we assumed that $\beta_1 > 0$,

$$\lim_{\rho_{xz}\to 0^+} \frac{\sqrt{\sigma_u^2+\delta^2\sigma_w^2}}{\beta_1\rho_{xz}\sigma_x\sqrt{n}} = +\infty$$
$$\lim_{\rho_{xz}\to 0^+} \lambda\left(\frac{\beta_1\rho_{xz}\sigma_x\sqrt{n}}{\sqrt{\sigma_u^2+\delta^2\sigma_w^2}} - z_\alpha\right) = \lambda(-z_\alpha) \in \mathbb{R}_+$$

Since the other components of Δ_{IV} are finite and do not depend on ρ_{xz}, $\lim_{\rho_{xz}\to 0^+} \Delta_{\text{IV}} = +\infty$. Since Δ_{IV} is a continuous function of ρ_{xz} on $(0,1]$, for any set of parameter values consistent with our hypotheses, $\exists \rho_{xz} \in (0,1]$ s.t. $\Delta_{\text{IV}} > 0$, *i.e.* s.t. $E_{\text{IV}} > E_{\text{OVB}}$. Since $\pi_1^2\sigma_z^2 = \rho_{xz}^2\sigma_x^2$, it is also straigthforward to show, for any set of parameter values consistent with our hypotheses, $\exists \pi_1 \in \mathbb{R}_+^*$ s.t. $E_{\text{IV}} > E_{\text{OVB}}$. □

We thus have shown that, <u>if the correlation between the instrument and the endogenous variable is low, the exaggeration bias of the IV can be larger than that of the "naive" regression, even in the presence of confounders</u>. The trade-off, *ie* the value of Δ, is affected by this correlation, or the strength of the instrument.

Yet, the value and the sign of Δ depend on the value of the other parameters in a non-trivial maner. In a concrete setting, it might therefore be difficult to identify how "low" the correlation has to be for Δ to be positive.

V. 2.2 Reduced form

Proposition 4. *Under the data generating process described in equation 1, for some parameter values, $E_{\text{RED}} > E_{\text{OVB}}$. More specifically, $\exists p_1 \in (0,1)$ s.t. $E_{\text{RED}} > E_{\text{OVB}}$.*

Proof. For clarity, we focus on a small number of treated observations, *i.e.* p_1 close to 0. The exact same reasoning applies for a small number of non-treated observations, *i.e.* p_1 close to 1 since $x \mapsto x(1-x)$ is symmetric around 0.5 and takes the value 0 in 0 and 1. Combining lemmas 2, 3 and 1, we get:

$$\Delta_{\text{RED}} = E_{\text{RED}} - E_{\text{OVB}} = \frac{\sqrt{(\delta+\beta_1\gamma)^2\sigma_w^2+\sigma_u^2+\beta_1^2\sigma_\epsilon^2}}{\beta_1\sqrt{n\times p_1(1-p_1)}}\lambda\left(\frac{\beta_1\sqrt{n\times p_1(1-p_1)}}{\sqrt{(\delta+\beta_1\gamma)^2\sigma_w^2+\sigma_u^2+\beta_1^2\sigma_\epsilon^2}}-z_\alpha\right)$$
$$-\frac{\sqrt{\sigma_u^2+\delta^2\sigma_w^2\left(1-\frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}}{\beta_1\sigma_x\sqrt{n}}\lambda\left(\frac{\sigma_x\sqrt{n}\left(\beta_1+\frac{\delta\gamma\sigma_w^2}{\sigma_x^2}\right)}{\sqrt{\sigma_u^2+\delta^2\sigma_w^2\left(1-\frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}}-z_\alpha\right)$$

The proof is analogous to that of proposition 3. We want to show that $\exists p_1 \in (0,1)$ s.t. $\Delta_{\text{RED}} > 0$. First, note that only E_{RED} depends on p_1. Since E_{RED} increases when p_1 gets further away from 0, Δ_{RED} also increases when $p_1 \to 0^+$ (or equivalently $p_1 \to 1^-$). Now, for fixed and finite values of the other parameters, and since we assumed that $\beta_1 > 0$,

$$\lim_{p_1\to 0^+}\frac{\sqrt{(\delta+\beta_1\gamma)^2\sigma_w^2+\sigma_u^2+\beta_1^2\sigma_\epsilon^2}}{\beta_1\sqrt{n\times p_1(1-p_1)}} = +\infty$$
$$\lim_{p_1\to 0^+}\lambda\left(\frac{\beta_1\sqrt{n\times p_1(1-p_1)}}{\sqrt{(\delta+\beta_1\gamma)^2\sigma_w^2+\sigma_u^2+\beta_1^2\sigma_\epsilon^2}}-z_\alpha\right) = \lambda(-z_\alpha) \in \mathbb{R}_+$$

Since the other components of Δ_{RED} are finite and do not depend on p_1, $\lim_{p_1\to 0^+}\Delta_{\text{RED}} = +\infty$ ($= lim_{p_1\to 1^-}\Delta_{\text{RED}}$). Since Δ_{RED} is a continuous function of p_1 on $(0,1)$, for any set of parameter values consistent with our hypotheses, $\exists p_1 \in (0,1)$ s.t. $\Delta_{\text{RED}} > 0$, *i.e.* s.t. $E_{\text{RED}} > E_{\text{OVB}}$. □

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